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Autonomous Navigation for Personal Mobility Vehicles Considering Passenger Tolerance to Approaching Pedestrians

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Abstract: There are high expectations for autonomous personal mobility vehicles (PMVs) to support the mobility of older people. Autonomous navigation systems are being developed to assist mobility in public areas with mixed pedestrian traffic, such as airports and shopping malls. For autonomous navigation of PMVs, achieving both comfort and efficiency, even in crowded environments, is important. In this study, we focused on the characteristic of passenger tolerance, in which a passenger's discomfort is relatively small concerning an approaching pedestrian. The objective was to propose an efficient autonomous navigation method without increasing passenger discomfort, considering the characteristics of passenger tolerance. First, the passenger tolerance characteristics were clarified through data analysis of a previous study's dataset and a newly collected dataset. Next, a path-planning method considering the characteristics was proposed, and the proposed method was evaluated by numerical simulations. The evaluation results showed that the proposed method has a potential to achieve efficient autonomous navigation in crowded environments.

Keywords: personal mobility vehicle; autonomous navigation; path planning; tolerance; passenger characteristics

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1. Introduction

Currently, automobiles play a crucial role in daily activities, such as shopping. However, accidents involving older drivers in Japan have become a social issue, leading some individuals to voluntarily give up their driver's licenses. Consequently, with the increasing population aging, there is a growing need for mobility support to assist the daily lives of older people. Therefore, personal mobility vehicles (PMVs), such as mobility scooters, are considered a means of mobility support. However, accidents resulting from user errors and misjudgments in operation have been reported [1]. Accordingly, developing autonomous functions independent of the users' abilities is crucial; ongoing efforts are being made in this direction [2–4].

In contemplating the societal implementation of autonomous PMVs in pedestrian spaces, it is important to consider not only the ability of the individual to reach their destination safely but to move efficiently and comfortably [5]. This efficiency can be interpreted as "short travel time to the destination". In a study investigating the travel time of autonomous vehicles, the "Freezing Robot Problem" was identified. This problem indicates the significant increase in travel time when mobility comes to a halt or takes extensive detours in crowded environments where a wide margin is maintained between pedestrians [6]. On the other hand, studies regarding the comfort of autonomous vehicles have investigated the comfort of both passengers and pedestrians in pedestrian spaces. Approaches to improve comfort focus on the relationship between passenger comfort and factors such as acceleration and jerk [7]. Furthermore, a concept exists known as personal

space (PS), where humans feel discomfort when that space is invaded by others. PS influences the comfort of passengers and pedestrians [8]. Based on this concept, previous studies have proposed methods to make autonomous movement comfortable for passengers and pedestrians [9,10].

There is a trade-off relationship between efficiency and comfort in crowded environments where pedestrians and PMVs approach each other. The PS concept suggests that PMVs should make significant avoidance behavior to obtain comfort. However, the larger the avoidance behavior, the longer the travel time. Considering this relationship, Harada et al. [11] explored effective avoidance strategies for autonomous PMVs from the perspectives of both pedestrians' comfort and PMV's travel time. However, to achieve socially acceptable movement, it is also essential to consider passenger comfort.

In crowded environments where the invasion of a passenger's PS by surrounding pedestrians is unavoidable, PMV passengers do not necessarily feel increased discomfort due to approaching pedestrians. Such instances can be viewed as situations where passengers tolerate invasion into their PS. If the characteristics of passenger tolerance (e.g., situations where passenger discomfort is relatively small concerning approaching pedestrians) are clarified, autonomous PMV movement satisfying both efficiency and comfort can be achieved. Therefore, this study aims to clarify the characteristics of passenger tolerance and propose a path-planning method considering the clarified characteristics, which can achieve efficient autonomous navigation without increasing passenger discomfort.

The remainder of this paper is organized as follows. Section 2 describes passenger tolerance to approaching pedestrians. Section 3 shows an investigation of its characteristics. Section 4 describes the proposed path-planning method considering the passenger tolerance characteristics. Section 5 shows the validation of the proposed method. Lastly, Section 6 summarizes the findings of this study.

2. Passenger Tolerance to Approaching Pedestrians

This study considers tolerance as a relative reduction of passenger discomfort towards approaching pedestrians in crowded environments. We investigated the study of Isono et al. [12], which discusses the comfort recognition of autonomous PMV passengers, to extract factors that impact a passenger's tolerance and to set the parameters for passenger tolerance in this study.

In [12], they constructed a comfort recognition model of an autonomous PMV passenger and proposed a novel path-planning method for autonomous PMV navigation considering the model. They applied Endsley's situation awareness (SA) model [13] to autonomous PMV passengers to describe their comfort recognition. Based on the SA model, passengers are assumed to perceive a situation in three levels: "Perception of the elements in the Current Situation", "Comprehension of the Current Situation", and "Projection of the Future Status". Their results revealed a strong positive correlation between the PS invasion ratio of passengers invaded by pedestrians and quantified passenger discomfort based on subjective assessments. The "PS invasion ratio" is an index to assess objective discomfort, as demonstrated in [9], which describes the extent to which a pedestrian invades a passenger's PS. Although there was a strong correlation, the results showed variations in passenger discomfort among scenes with similar PS invasion ratios. From the levels of the SA model, the PS invasion ratio corresponds to "Comprehension of the Current Situation". Thus, it can be suggested that factors related to "Projection of the Future Status" may cause variations in passenger discomfort.

From the above, "Projection of the Future Status" factors were extracted as factors impacting passenger tolerance. Therefore, in this study, scenes with small discomfort among the scenes with varied discomfort and a similar PS invasion ratio are focused on, and relatively small discomfort among similar PS invasion ratios due to "Projection of the Future Status" is set as passenger's tolerance.

3. Characteristics of Passenger Tolerance to Approaching Pedestrians

This section investigates characteristics of passenger tolerance to approaching pedestrians in two steps. Firstly, the dataset of Isono et al. [12] was analyzed to identify factors affecting passenger tolerance. Secondly, a participant experiment was conducted to understand their influence, overcoming a limitation of the former analysis, and the obtained dataset was analyzed. In this procedure, Pearson correlation coefficients were analyzed between each parameter and passenger discomfort or tolerance to investigate the relationship between these variables. A significant correlation was considered to be present when the p-value was less than 0.05.

3.1. Analysis of Previous Study's Dataset

To identify the potential factors influencing passenger tolerance to approaching pedestrians in crowded pedestrian spaces, the dataset collected in a previous study [12] was adopted. This dataset includes comfort evaluation data of PMV passengers obtained through a virtual reality (VR) simulator experiment. Utilizing the dataset, first, an index to assess passenger tolerance was set. Next, factors affecting passenger tolerance were specified.

3.1.1. Previous Study's Dataset

An autonomous PMV navigating through a crowded environment in a shopping mall was simulated and reproduced in a VR environment, as shown in Figure 1. The PMV navigated itself inside a 6-m wide pathway while avoiding 30 pedestrians coming from the opposite direction (pedestrian density approximately $0.125/\mathrm{m}^2$). The PMV moved based on a pedestrian avoidance method considering passenger comfort [14], while pedestrians moved based on a social force model (SFM), considering the closest approach to surrounding pedestrians [15]. Sixteen scenarios with different movements of both the PMV and the pedestrians were set.

Figure 1. Sample image of the virtual reality environment.

Figure 2 shows the experimental setup. The participants were presented with each scenario reproduced in a VR environment using a head-mounted display (HMD). During the playback of each scenario, the participants were instructed to grip a lever when they felt uncomfortable and to adjust the grip according to the degree of discomfort. This procedure enables the collection of passenger discomfort along the time axis. Ten healthy male participants (23.0 ± 0.77 years old) participated in this experiment.

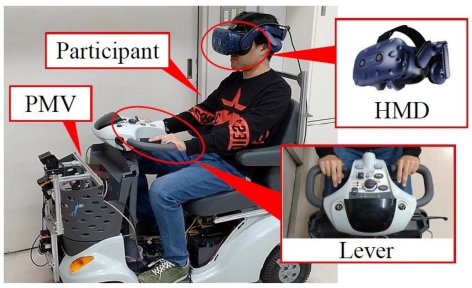


Figure 2. Experimental setup.

3.1.2. Passenger Tolerance Metric

The relationship between the PS invasion ratio and passenger discomfort was investigated to examine a passenger tolerance index. In the previous research [12], a strong linear relationship was found between discomfort and the ratio. The maximum PS invasion ratio in PS invasion scenes was adopted as an index of the PS invasion ratio. The shape of the passenger's PS, referenced from previous studies [9,12], was defined as an ellipse (Figure 3) with dimensions considering the size of the PMV as l_f = 5.62 m and l_s = 0.80 m. The PS invasion ratio I is calculated as shown in Figure 3. Regarding passenger discomfort, the maximum value of passenger discomfort in PS invasion scenes was adopted. The PS invasion scene is defined as the duration from when a pedestrian enters the passenger's PS until it exits it.

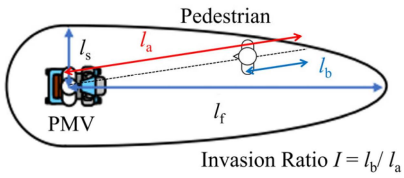


Figure 3. Definition of personal space (PS) and the PS invasion ratio.

The result of the relationship between the PS invasion ratio and passenger discomfort revealed a proportional relationship (Figure 4). The correlation coefficient between the two indices was significant and high (r = 0.81, p < 0.01). An approximation line was set as the criterion for passenger discomfort concerning the PS invasion ratio to evaluate the variability of passenger discomfort for a similar level of PS invasion ratio. While various monotonically increasing functions can be considered to represent the discomfort of passengers with respect to the PS invasion ratio, the straight approximation line simply calculated by the least squares was adopted. This approach ignores complex variations caused by various factors, focusing on capturing the primary trend in the data to identify factors that affect the variations significantly. Then, the tolerance metric q is defined as Equation (1) and Figure 4 based on the line; s denotes the passenger discomfort, and s_0 denotes the discomfort obtained from the approximation line.

$$q = s_0 - s \tag{1}$$

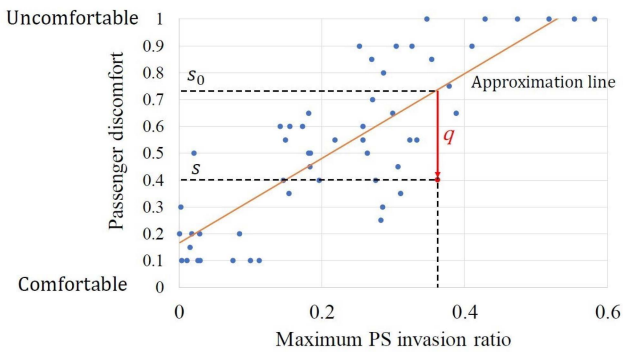


Figure 4. Passenger discomfort as a function of the PS invasion ratio.

3.1.3. Factors Affecting Passenger Tolerance

From the result of Section 2, the factors related to "Projection of the Future Status" are suggested to affect passenger tolerance. Therefore, two factors related to a passenger's predictions of discomfort due to the approaching pedestrian were focused on. One factor is the lateral distance between the passenger and the pedestrian when it passes by the PMV. If this is predicted to be long, the passenger can pass by the pedestrian with enough margin and low discomfort. The other factor is the magnitude of the pedestrian's or the

PMV's avoidance behavior toward each other. If this avoidance is predicted to be large, the passenger can pass by the pedestrian with enough margin with low discomfort. The relationships between the two factors and the tolerance metric q were investigated to examine whether the above factors affect passenger tolerance.

Indices representing the two factors were set to investigate the relationships between the two factors and passenger tolerance. The indices were calculated in the vehicle relative coordinate system. The predicted pedestrian behavior in this coordinate system was assumed to be a circular turn with a constant curvature radius, as shown in Figure 5. The predicted pedestrian behavior was calculated based on the position and velocity of the PMV and the pedestrian before the pedestrian invades the passenger's PS. For the first factor, the predicted passing distance h was set as the distance between the passenger and the intersection point of the predicted pedestrian trajectory and the rear-wheel axle of the PMV. The curvature radius R of the predicted pedestrian trajectory was set for the second factor. A smaller R implies a larger change in the pedestrian or the PMV speed, indicating larger avoidance actions.

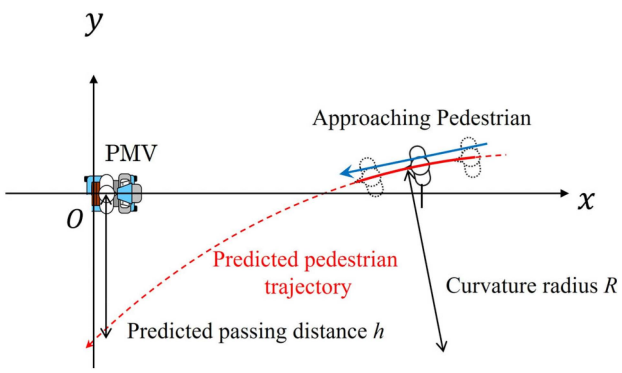


Figure 5. Definition of the predicted passing distance h and the curvature radius R of the predicted trajectory.

The predicted passing distance h was calculated for each PS invasion scene, and the relationship between h and the tolerance metric q was investigated. As shown in Figure 6, the relationship between h, calculated when the pedestrian was 6.7 m ahead of the PMV, and q showed a significant positive correlation ($r = 0.58$, $p < 0.05$). This result suggests that a larger h makes it easier for passengers to predict that there will be no excessively close approach or collision with the pedestrian, allowing them to tolerate the pedestrian's approach.

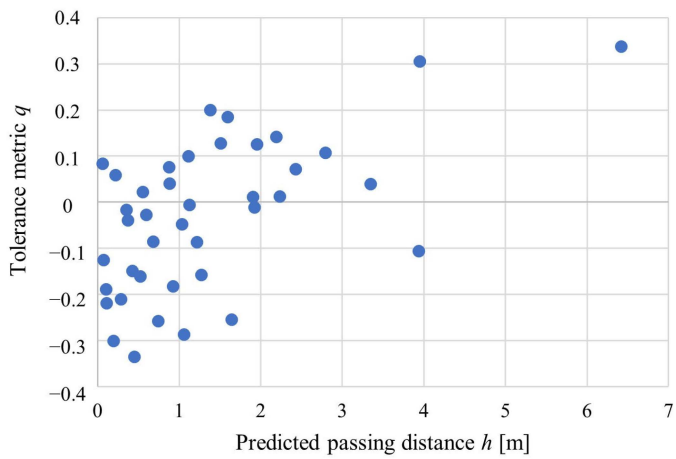


Figure 6. Relationship between the predicted passing distance h and the tolerance metric q.

As for the curvature radius R, the minimum value among the pedestrian conditions in the range 6.0–7.0 m ahead of the PMV was calculated to consider the largest avoidance

behavior. As shown in Figure 7, the relationship between R and the tolerance metric q showed a significant negative correlation ($r = -0.71$, $p < 0.05$). This result indicates that a smaller R makes it easier for passengers to tolerate the situation. Furthermore, it can be suggested that passengers are more likely to tolerate pedestrian approaches when the curvature radius of the pedestrian's trajectory is small because they can predict significant avoidance movement by either the pedestrian or the PMV and anticipate no collision.

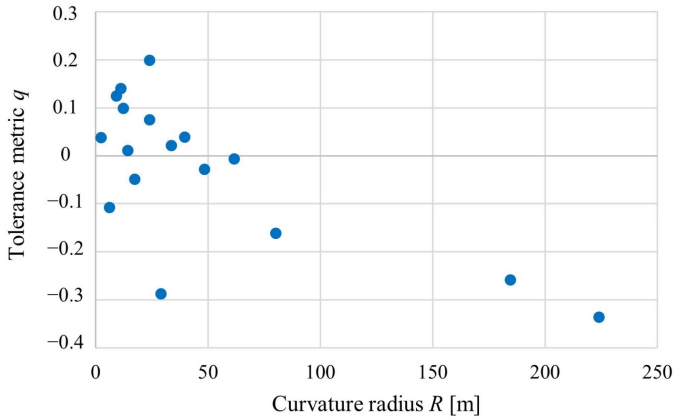


Figure 7. Relationship between the curvature radius R and the tolerance metric q.

From the previous study's dataset analysis, it was suggested that the predicted passing distance and the curvature radius of the predicted pedestrian trajectory affect passenger tolerance. In this dataset, the autonomous PMV actively avoided the surrounding pedestrians. By contrast, suppressing avoidance behavior without compromising passenger discomfort is required to achieve efficient and comfortable movement. Thus, it is necessary to clarify the characteristics of passenger tolerance regarding the factors in cases when the PMV does not perform avoidance behaviors.

3.2. Collection and Analysis of a Novel Experiment Dataset

Similar to the previous study's dataset, a VR simulator experiment was conducted to collect passenger discomfort data when the autonomous PMV did not perform avoidance actions, ensuring safety. With the collected data, the two factors assumed to affect passenger tolerance were investigated similar to the previous section.

3.2.1. Experimental Conditions

In this experiment, scenarios in which the PMV moves in a crowded environment at a constant speed were simulated and reproduced in the VR environment. The PMV speed was set to 1.1 m/s, and the pedestrian density and aisle shape were identical to the previous study's dataset. Additionally, for each scenario, one pedestrian invaded the passenger's PS along a predefined path, as shown in Figure 8. The pedestrian approached the PMV at a constant speed, initiated a turn at a distance of d from the PMV front, and entered the passenger's PS while maintaining a constant curvature and speed in the vehicle relative coordinate system. After leaving the PS, the pedestrian moved away from the PMV in a straight line and at a constant speed. The other pedestrians moved according to an SFM model [15] in the opposite direction of the PMV.

To clarify the positive or negative impact of the predicted passing distance h and the curvature radius R of the predicted pedestrian trajectory on passenger tolerance, a total of nine conditions with three levels of h (0.7, 0.9, and 1.3 m) and R (30, 60, and 90 m) were set, as shown in Table 1. These conditions for h and R were set as values within the range where the PS invasion scenarios are frequently distributed, as shown in Figures 6 and 7, to investigate the characteristics in the targeted crowded environments. The maximum PS invasion ratio and the distance where the pedestrian starts to turn were set constant, respectively ($I = 0.25$, $d = 7.5$ m).

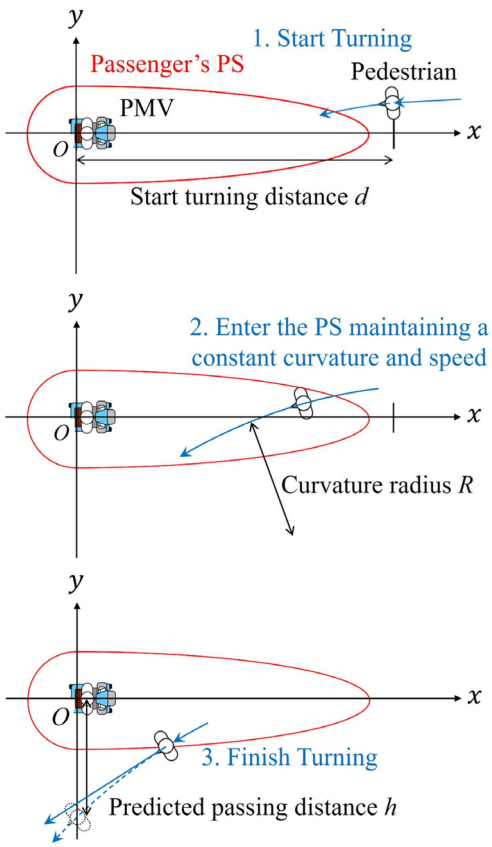


Figure 8. Pedestrian trajectory parameters.

Table 1. Experimental conditions.

Condition ID	Predicted Passing Distance h (m)	Curvature Radius R (m)
1	0.7	30
2	0.7	60
3	0.7	90
4	0.9	30
5	0.9	60
6	0.9	90
7	1.3	30
8	1.3	60
9	1.3	90

3.2.2. Experimental Method

With the conditions mentioned in the previous section, nine scenarios were created. The experiment was conducted in the same methodology as the previous study [12] for each scenario. Time-series data of passenger discomfort on a scale of 0 to 1 was obtained. In this experiment, the maximum PS invasion ratio was set constant. Therefore, the criterion for passenger discomfort due to the PS invasion ratio was set to a constant value, specifically the median of the discomfort scale. Tolerance was then evaluated as the relative discomfort compared to this constant value. Thus, 0.5 was adopted as the median value of the discomfort scale, and the tolerance metric $q\prime$ was calculated as the relatively small discomfort with the passenger discomfort value s'(Equation (2)).

$$q' = 0.5 - s' \tag{2}$$

Eleven healthy male and female participants (22.8 ± 2.29 years old) participated in this experiment. Before the experiment, participants were provided with an explanation of the procedures, and informed consent was obtained. This study was conducted under the approval of the Ethics Review Committee of the University of Tokyo Life Sciences Committee.

3.2.3. Experimental Results

It was clarified that two factors influence passenger tolerance, as shown in Figures 9 and 10. The relationship between the predicted passing distance h and the tolerance metric $q\prime$ showed a similar trend with the previous study's dataset (Figure 9). A larger h allowed the passengers to tolerate the pedestrians invading their PS. By contrast, the relationship between curvature radius R and $q\prime$, as illustrated in Figure 10, showed an opposite tendency. A larger R (i.e., a straighter pedestrian trajectory) makes the passenger tolerate the pedestrian's approach. This result suggests that, when pedestrians move in a straight line, it is easier for the passengers to judge their moving directions and predict their future trajectories relative to themselves. If the pedestrians are not predicted to come close, the concerns about potential collisions with them can be reduced.

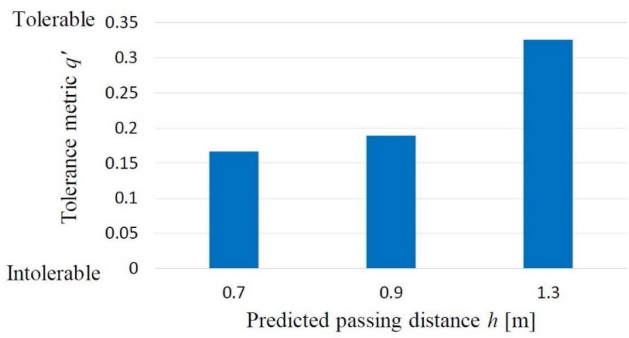


Figure 9. Impact of the predicted passing distance h on the tolerance metric $q\prime$.

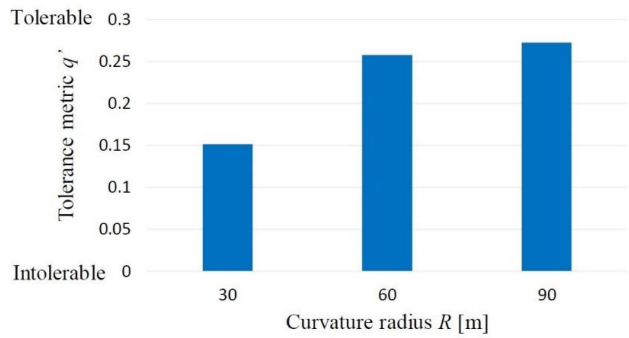


Figure 10. Impact of the curvature radius R on the tolerance metric $q\prime$.

Based on Sections 3.1 and 3.2 analysis results, the characteristics of PMV passenger tolerance to approaching pedestrians in crowded pedestrian spaces can be summarized as follows:

- When the predicted passing distance h is larger, the passenger tolerates the pedestrian's approach.
- When the vehicle does not perform active avoidance behavior, the passenger tolerates the pedestrian's approach when the curvature radius R is larger.

3.3. Discussion on the Characteristics of Passenger Tolerance

The relationship between the curvature radius R of the predicted pedestrian's trajectory and the passenger tolerance exhibited different trends in the cases where the PMV performed avoidance behavior (Section 3.1) and the cases where the PMV did not (Section 3.2). The curvature radius R is influenced by both the behavior of the pedestrian

and the PMV. In the scenario outlined in Section 3.2, since the PMV moves at a constant speed and direction, the relative sizes of the curvature radius calculated in the vehicle relative coordinate system will be the same as in the absolute coordinate system when comparing different PS intrusion scenarios. On the other hand, in the scenario outlined in Section 3.1, if a pedestrian moves linearly in the absolute coordinate system but the PMV experiences significant speed changes, the pedestrian's behavior will appear more pronounced in the vehicle relative coordinate system, resulting in a smaller radius of curvature. When the PMV is avoiding an approaching pedestrian, as in the case in Section 3.1, the passenger will be relieved if the PMV is making a large avoidance behavior (small curvature) because it can be anticipated that a collision with the pedestrian will be avoided. By contrast, when the pedestrian is avoiding the PMV and the PMV is moving in a straight line, as in the case in Section 3.2, the passenger will be relieved if the pedestrian is moving in a straight line (large curvature) because it is easier for the passenger to predict the pedestrian's behavior and anticipate that no collision will occur. Therefore, it is assumed that this difference in the scenarios led to the different trends.

However, in the experiments described in Sections 3.1 and 3.2, the sample size is insufficient to conclude that the identified characteristics represent general characteristics. In addition, the simulations in VR environments have limitations, such as the inability to fully replicate scenarios with the unpredictability of pedestrian movement in real-world environments. In real-world environments, pedestrians move complexly in response to various factors, such as their personalities and situations, while avoiding collisions with others, which could potentially influence passenger tolerance. Passengers may also find it more difficult to perceive their sense of distance from surrounding pedestrians. In Sections 3.1 and 3.2, primary trends were analyzed as an initial evaluation in safe experimental environments. Therefore, future experiments should be conducted in real-world environments with safety conditions to comprehensively and certainly understand these characteristics.

4. Path-Planning Method Considering Passenger Tolerance Characteristics

This section proposes a path-planning method considering passenger tolerance characteristics toward approaching pedestrians. First, prerequisites and required functions were set based on the previous section's results. Next, a path-planning method satisfying the prerequisites and required functions was proposed.

4.1. Prerequisites and Required Functions

The suppression of avoidance behavior for pedestrians for whom the PMV passenger can tolerate the invasion of PS was focused on to achieve efficient autonomous mobility in crowded environments. Thus, a PMV's path-planning method that suppresses its avoidance behavior when the passenger can tolerate approaching pedestrians was considered based on passenger tolerance characteristics. This method emphasizes avoidance for pedestrians that the passenger cannot tolerate to increase passenger discomfort. By contrast, when a tolerable pedestrian is approaching, the method moves the PMV in a straight line toward its destination. The PMV autonomous mobility system was assumed to consist of self-localization and environment recognition using LiDAR sensors, etc., and path planning to the destination. The PMV assumed in this method was a mobility scooter, which is a non-holonomic mobility system. It was also assumed that all surrounding pedestrians could be recognized with the perfect accuracy of their position and velocity, and there are no computational resource limitations.

From the results of Section 3.3, it was clarified that a passenger can tolerate a pedestrian's invasion of their PS when the pedestrian's avoidance behavior in the range of 6.0–7.0 m ahead of the PMV is small, and the predicted passing distance is long. Based on these characteristics, the required function was set to suppress the avoidance behavior of the PMV for tolerable pedestrians with small avoidance behavior just before invading their PS, and long predicted passing distances to select a path with a shorter travel time.

4.2. Proposal of a Path-Planning Method

The proposal is based on the path-planning method proposed in the previous study [12]. The existing method [12] consists of three path planners: the global path planner, the middle path planner, and the local path planner. As Figure 11 shows, the middle path planner outputs intermediate target paths by utilizing the global path to the destination the global path planner provides. This planner considers future interactions with pedestrians based on the positional information of the PMV and the recognized pedestrians, along with map data. Subsequently, the local path planner considers the intermediate target paths generated by the middle path planner to produce the final control values for the PMV, namely, the target speed v and steering angle δ. The middle path planner plays a role in proactively preventing pedestrians within a mid-range distance of up to 10 m from the vehicle from entering the PS. The local path planner, on the other hand, is responsible for avoiding closer pedestrians within a range of 5 m. Therefore, in the proposed method, the local path planner is improved to suppress avoidance behavior for tolerable pedestrians, while the functions of the middle path planner and global path planner remain unchanged.

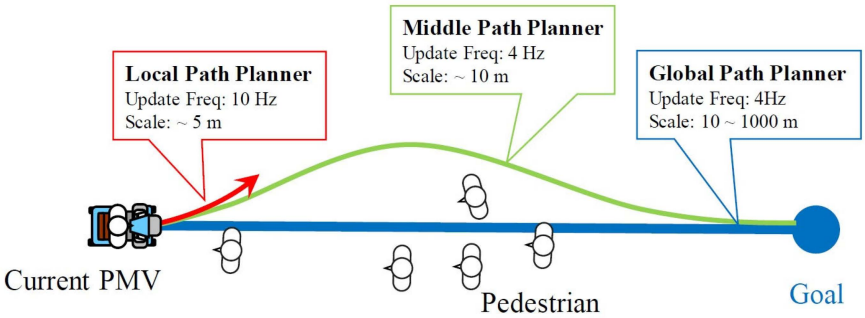


Figure 11. Paths generated by 3-layer path planners.

In [12], the local path planner outputs a combination of v and δ, maximizing the evaluation function in Equation (3). L_{goal} denotes the distance to approach the intermediate target path within an evaluation time step T_{local}, L_{inv} denotes the PS invasion ratio of the surrounding pedestrians, and β_{goal} and β_{inv} denote the weighting constants of L_{goal} and L_{inv}, respectively. L_{inv} is expressed by Equation (4), incorporating the PS invasion ratios $I_{wall}(t)$ from static obstacles, such as walls, and $I_{ped}(t)$ from surrounding pedestrians, acting as a penalty item that decreases as the PS invasion ratio increases. Additionally, in [12], the PS of a passenger is constant, and Equation (5) is used to calculate the sum of the PS invasion ratio for each pedestrian, where $I_j(t)$ and $I_k(t)$ represent the PS invasion ratios of pedestrians j and k, respectively. This study proposes an improved path-planning method by refining this local path planner.

$$G(v, \delta) = \beta_{goal}\, L_{goal} + \beta_{inv} L_{inv} \tag{3}$$

$$L_{inv} = -\int_0^{T_{local}} (I_{wall}(t) + I_{ped}(t))dt \tag{4}$$

$$I_{ped}(t) = \sum\nolimits_{j \in \{k | I_k(t) > 0\}} I_j(t) \tag{5}$$

In the proposed method, the evaluation of the PS invasion ratio is reduced for the pedestrians for whom the passenger can tolerate, compared to other pedestrians. This function is achieved by decreasing the penalty in L_{inv}, thereby suppressing the PMV's avoidance behavior. The PS size of the passenger is changed for pedestrians for whom the passenger can tolerate to reduce the PS invasion ratio evaluation. Specifically, the size of the PS in the forward direction is reduced based on the behavioral history of the target pedestrian to make the PS invasion ratio smaller than it is. Based on the analysis results, the size of the PS in the forward direction is calculated using the minimum curvature

radius R_{min} of the predicted pedestrian's trajectory among the range where the pedestrian is 6.0–7.0 m ahead of the PMV, as shown in Equation (6). Here, γ_R is a constant satisfying Equation (7), and a threshold of 0.5 is set for the rate of PS size reduction to prevent collisions between the PMV and the pedestrians caused by an excessively small PS. As depicted in Figure 12, the larger the R, the shorter $l_f\prime$, thereby reducing the PS invasion ratio, as expressed in Equation (8).

$$l'_f = max\left(0.5,\ {\gamma_R}^{R_{min}}\right) \cdot l_f \tag{6}$$

$$0 < \gamma_R < 1 \tag{7}$$

$$\frac{l'_b}{l'_a} < \frac{l_b}{l_a} \tag{8}$$

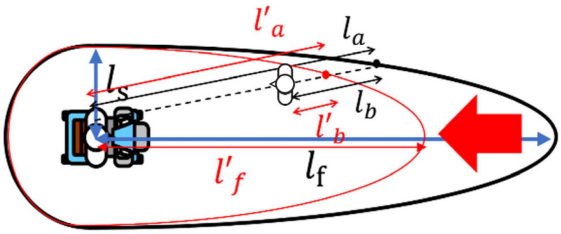


Figure 12. Scale reduction of PS for tolerable pedestrians.

For pedestrians whose proximity is tolerable, the lateral size of the passenger's PS can also be reduced, making the path straighter. However, if the PS is reduced in the lateral direction, the PMV may suppress avoidance behavior toward pedestrians whose predicted passing distance is short. This is contrary to the passenger's tolerance characteristic of being able to tolerate an approaching pedestrian with a large predicted passing distance. Thus, the size of the PS was reduced only in the forward direction.

5. Validation of Proposed Method

Evaluation metrics were established and used to compare the proposed method with the existing method outlined in [12] through numerical simulations conducted in environments similar to those targeted by [12]. This was conducted to confirm that the method proposed in Section 4 satisfies the functional requirements specified in Section 4.1, and to accurately evaluate the impact of adding the new function on the efficiency of autonomous navigation through fair comparisons. In the comparison of the methods, t-tests were conducted under the same pedestrian density conditions, and a significant difference was identified when the p-value was less than 0.05.

5.1. Evaluation Method

In this study, some crowded environments similar to the simulation in [12], where it is challenging for the PMV to reach the destination efficiently due to avoidance behavior, were focused on. Therefore, numerical simulations were conducted in a corridor, considered sufficiently wide for the vehicle to execute avoidance behaviors, where numerous pedestrians approach the vehicle from the front.

In the simulation, as shown in Figure 13, the width and length of the pathway were 6.0 m and 40 m, respectively. The PMV was controlled by either the proposed method or the existing method. The PMV started from $(x,\ y) = (0,\ 1)$ heading for $(x,\ y) = (40, -1)$, which is assumed to be a waypoint on the global path. A trial of autonomous navigation was considered complete if the PMV reached the goal line $(x = 32.5\ \text{m})$ within 60 s after starting. Pedestrians were assigned random initial positions, velocities, and destinations and moved according to an SFM model [15]. Regarding the path-planning method, a parameter study was conducted in the environment for the weight constants β_{goal} and β_{inv}. Then, the combination of the constants was set $\left(2^{-5}, 2^{-5}\right)$, a combination optimal from the

perspective of passenger comfort, which was emphasized in [12]. A constant value of 0.995 was adopted for γ_R as the value that satisfies Equation (7) in Section 4.2.

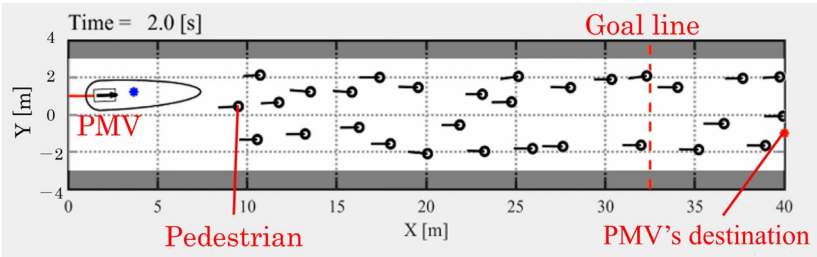


Figure 13. Overviews of numerical simulation environment.

To investigate the differences in the behavior between the existing and proposed methods due to differences in pedestrian density, five conditions with different numbers of pedestrians (10, 20, 30, 40, and 50) were set. Additionally, one simulation trial was conducted for the condition with no pedestrians. Thus, this simulation's pedestrian density ranged from approximately 0.0 to $0.2/\mathrm{m}^2$. One thousand simulation trials were conducted for each pedestrian density condition with different initial conditions. The same pedestrian initial conditions were applied to both existing and proposed methods.

5.2. Evaluation Metrics

Evaluation metrics were set in two perspectives. One was to verify whether the method proposed in Section 4.2 meets the requirement of suppressing avoidance behavior for approaching pedestrians tolerated by the passenger, introduced in Section 4.1. The duration of the PMV executing avoidance behavior during autonomous navigation was evaluated from this perspective. The PS invasion ratios of tolerable and intolerable pedestrians were also compared to confirm the suppression for not intolerable but tolerable pedestrians. The other was to verify whether the proposed method can achieve efficient autonomous movement without increasing passenger discomfort. In addition to passenger discomfort estimation, the travel time of autonomous navigation was evaluated from this perspective.

For the first evaluation perspective, the avoidance duration was defined based on the definition of avoidance behavior in the previous study [12]. The previous study considered deceleration and turning actions as avoidance. Therefore, actions were defined as deceleration when they satisfied Equation (10), indicating a slow speed significantly lower than the maximum speed v_{max}. Similarly, actions were defined as turning when they satisfied Equation (11), indicating either a large steering angle δ [degree] for continuous turning or a large angular velocity indicating an attempt to start turning. The time the PMV performed avoidance behavior was defined as the duration that the PMV motion satisfied either Equation (9) or Equation (10).

$$v \leq 0.9 v_{max} \ \wedge \ \frac{\Delta v}{\Delta t} \leq 0 \tag{9}$$

$$|\delta| \geq 2.0 \ \vee \ \left|\frac{\Delta \delta}{\Delta t}\right| \geq 2.5 \tag{10}$$

The conditions of tolerable and intolerable pedestrian proximity were also defined. Considering that both the predicted passing distance h and the curvature radius R of the predicted pedestrian trajectory contribute to passenger tolerance, scenes satisfying the conditions in Equation (9) were considered scenes where the pedestrian's approach is tolerable; otherwise, the approach was deemed intolerable. To determine the calculation method of the predicted passing distance h, the correlation between h, calculated in different conditions where the pedestrian was 5.5–7.0 m ahead of the PMV, and q were investigated using the dataset in [12]. As a result, the strongest correlation was observed in the condition of 6.7 m, and significant correlations were also confirmed in the surrounding

range, as shown in Figure 14. It was therefore presumed that the passengers tended to make "Projection of the Future Status" and tolerate some approaching pedestrians around the time when the pedestrians were in this region. Thus, the passing prediction distance h in Equation (9) was calculated when a pedestrian was 6.7 m ahead of the PMV, which showed the strongest correlation. For the curvature radius R, the minimum value among the range 6.0–7.0 m ahead of the PMV was adopted. The time-averaged PS invasion ratio was then calculated separately for each pedestrian category.

$$h \geq 0.80 \ \wedge \ R \geq 100 \tag{11}$$

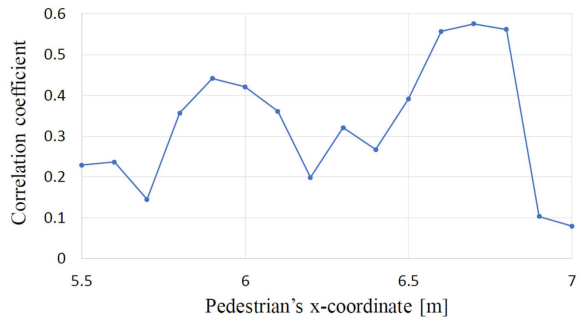


Figure 14. Relationship between the correlation and the pedestrian position at the prediction timing.

For the second evaluation perspective, passenger discomfort was estimated based on the PS invasion ratio of the surrounding pedestrians that the passenger cannot tolerate. A smaller time-averaged PS invasion ratio of the pedestrian deemed intolerable indicates that the method does not increase the PMV passenger's discomfort. Regarding the metric for efficiency, travel time was defined as the elapsed time from the start to the completion of the PMV's movement.

5.3. Evaluation Results

The results of the proposed method were compared with the existing method using the evaluation metrics set in Section 5.2 for each pedestrian density condition.

5.3.1. Avoidance Duration

As the pedestrian density increases, the avoidance duration becomes longer to prevent collisions and PS invasions by surrounding pedestrians (Figure 15). Therefore, it can be observed that the scenes where the proposed method should suppress the PMV's avoidance behavior are more likely in higher pedestrian density conditions.

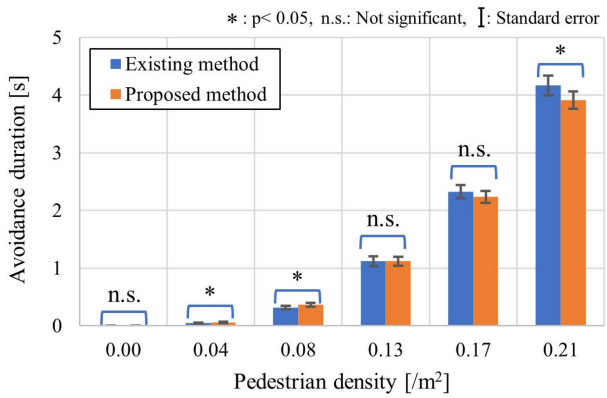


Figure 15. Comparison of avoidance duration among different pedestrian densities.

In the condition with 10 pedestrians (pedestrian density 0.04/m^2) and 20 pedestrians (pedestrian density 0.08/m^2), the average avoidance duration per simulation trial was significantly longer for the proposed method compared to the existing method ($p < 0.05$). On the other hand, in higher pedestrian density conditions, the proposed method's avoidance duration is shorter. Specifically, in the condition with 50 pedestrians (pedestrian density 0.21/m^2), the duration was significantly shorter for the proposed method compared to the existing method ($p < 0.05$). This result suggests that the proposed method can suppress avoidance, especially in situations where the avoidance duration becomes longer due to high pedestrian density.

5.3.2. Personal Space Invasion Ratio

The PS invasion ratio was compared between the two methods in two pedestrian categories: tolerable and intolerable pedestrians. To investigate the difference in the invasion ratios between tolerable and intolerable pedestrians with each method, the relative metric $I_{relative}$ calculated using Equation (12) with PS invasion ratios of tolerable and intolerable pedestrians was evaluated.

$$I_{relative} = I_{tolerable} - I_{intolerable} \tag{12}$$

$I_{tolerable}$ and $I_{intolerable}$ denote the time-averaged PS invasion ratios of tolerable and intolerable pedestrians, respectively. As shown in Figure 16, the invasion ratio of tolerable pedestrians is generally smaller than that of intolerable pedestrians, and the difference increases as pedestrian density grows. This is because intolerable pedestrians have a smaller h and approach the PMV more closely, and as pedestrian density increases, more pedestrians invade the PS.

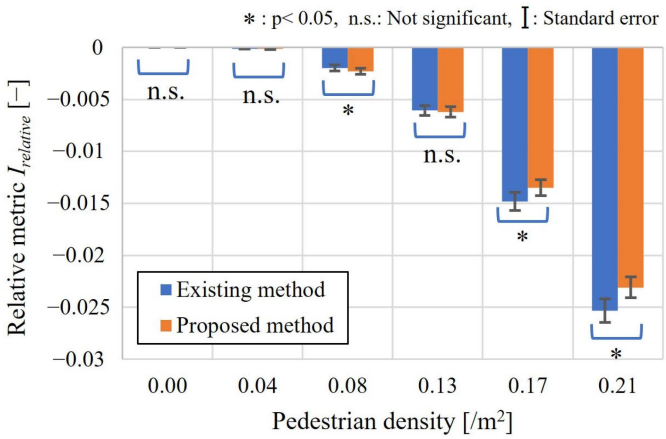


Figure 16. Comparison of the relative metric regarding the PS invasion ratios of tolerable and intolerable pedestrians.

For the perspective of examining whether the proposed method satisfies the required function set in Section 4.1, in the condition with 20 pedestrians (pedestrian density 0.08/m^2), $I_{relative}$ was significantly smaller for the proposed method compared to the existing method ($p < 0.05$). On the other hand, in higher pedestrian density conditions, specifically in the condition with 40 pedestrians (pedestrian density 0.17/m^2) and with 50 pedestrians (pedestrian density 0.21/m^2), the relative metric was significantly larger for the proposed method compared to the existing method ($p < 0.05$). This trend of the increasing metric, especially in higher pedestrian density conditions, suggests the PS invasion ratio tends to increase more for tolerable pedestrians than for intolerable pedestrians in autonomous navigation with the proposed method. Accordingly, the proposed method can suppress avoidance for tolerable pedestrians more likely than intolerable pedestrians in high pedestrian density environments.

The time-averaged PS invasion ratio of intolerable pedestrians $I_{intolerable}$ was also evaluated to estimate passenger discomfort during autonomous navigation of the existing

and proposed methods. As shown in Figure 17, the proposed method demonstrated smaller values than the existing method, particularly in high pedestrian density scenarios. In the condition with 40 pedestrians (pedestrian density 0.17/m^2), the PS invasion ratio was 0.015 for the existing method and 0.014 for the proposed method. Furthermore, in the condition with 50 pedestrians (pedestrian density 0.21/m^2), the ratio was 0.026 for the existing method and 0.024 for the proposed method. The PS invasion ratio of the proposed method was significantly smaller than that of the existing method ($p < 0.05$). Based on the positive correlation between the PS invasion ratio of surrounding pedestrians and passenger discomfort [12], this result suggests that the discomfort caused by intolerable pedestrians can be reduced. Therefore, it can be considered that the proposed method does not increase passenger discomfort.

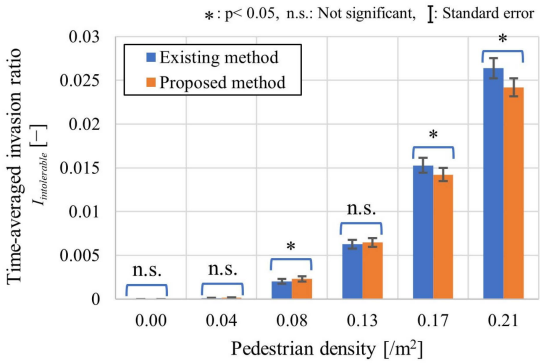


Figure 17. Comparison of the time-averaged PS invasion ratio of intolerable pedestrians.

5.3.3. Travel Time

Travel times for both methods were longer when the pedestrian density was higher, as shown in Figure 18. In the condition with 20 pedestrians (pedestrian density 0.08/m^2), the average travel time was also significantly longer for the proposed method compared to the existing method ($p < 0.01$). On the other hand, in higher pedestrian density conditions, the travel time of the proposed method was shorter compared to the existing method. Particularly, in the condition of 50 pedestrians (pedestrian density 0.21/m^2), the average travel time for the existing method was 33.1 s. By contrast, it was 32.8 s for the proposed method, and this difference between the methods was significant ($p < 0.01$). In the scenario with no pedestrians, the PMV takes 29.2 s to reach the destination. Thus, this time can be taken as the baseline, which any pedestrian avoidance strategy cannot further reduce. Among the highest pedestrian density conditions, the existing method resulted in a 3.9-s increase in travel time from the baseline, and the proposed method resulted in a 3.6-s increase. From the comparison of the increased time from the baseline due to pedestrian avoidance among the two methods, the proposed method reduced the increased time from the baseline by approximately 8%, which can be regarded as practically significant.

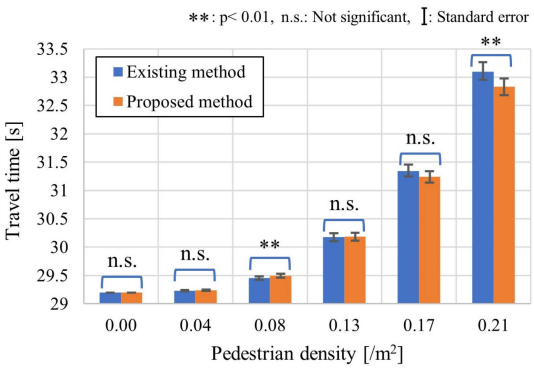


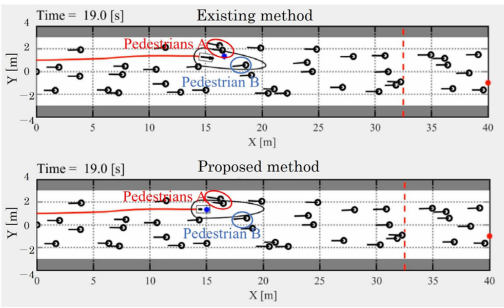
Figure 18. Comparison of the travel time among different pedestrian densities.

The numerical simulation results showed that the proposed method satisfies the required function of suppressing the avoidance behavior of the PMV for tolerable pedestrians, as set in Section 4.2. It can shorten the travel time without increasing passenger discomfort of autonomous navigation, particularly in crowded environments with high pedestrian density.

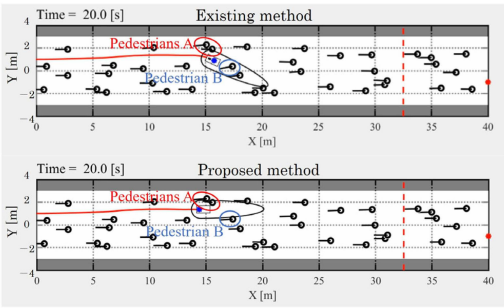
5.4. Discussion on the Proposed Method

Results showed that the proposed method has the potential to demonstrate efficient autonomous mobility without increasing passenger discomfort compared to the existing method in crowded scenarios with a pedestrian density of $0.15/\mathrm{m}^2$ or higher. This result is likely due to the increased occurrence of scenes where the passenger can tolerate invasion as the pedestrian density rises, providing more opportunities for the PMV to suppress avoidance behavior.

Particularly, comparing the paths generated by each method in a trial with 50 pedestrians (pedestrian density $0.21/\mathrm{m}^2$) revealed a significant difference in the chosen directions, as shown in Figure 19. In this scene, pedestrians A and B approached the PMV. Pedestrian A, walking along the wall, moved more linearly than pedestrian B, who was walking near the pathway center, as depicted in Figure 19a. The autonomous PMV with the existing method steers toward the direction of pedestrian B to reduce the number of PS invasions. By contrast, the autonomous PMV with the proposed method tolerated the approach of pedestrian A and suppressed the avoidance behavior. Consequently, 1 s later, as depicted in Figure 19b, the autonomous PMV with the existing method moved toward the pathway center, obstructing the pedestrian flow, and leading to a substantial increase in the PS invasion ratio and travel time compared to the proposed method. The existing method makes avoidance actions for all approaching pedestrians, leading to results where the PMV heads toward the pathway center to reach the destination and gets sandwiched by numerous pedestrians on both sides. On the other hand, the proposed method suppresses avoidance behavior to the approach of pedestrians moving in a straight line, particularly for those who are moving in a straight path in a streamlined pedestrian flow. In contrast to the existing method, the proposed method steers toward pedestrians moving in a straight line, especially those walking along the wall, aiming to maintain a straight trajectory without opposing the pedestrian flow, as shown in the lower figure of Figure 19a. This behavior is assumed to have contributed an efficient autonomous navigation in high pedestrian density environments.



(a) A scene of pedestrians approaching PMV



(b) The scene 1 second after the scene (a)

Figure 19. Comparison of PMV behavior between the existing and proposed methods in two scenes 1 s apart.

As the introduction mentions, safety is also a crucial requirement for an autonomous PMV. The collision count between the PMV and pedestrians per simulation trial was compared between the existing and proposed methods. As illustrated in Figure 20, the proposed method showed less collision count than the existing method in environments of higher pedestrian density. This difference is attributed to the fact that the proposed method exhibits fewer abrupt changes in behavior to the pedestrian flow. By following a trajectory closer to the wall, the proposed method can decrease the number of instances pedestrians invade the PMV's path. Thus, it is suggested that the proposed method can also contribute to the improvement as well as the achievement of efficient autonomous navigation in crowded environments with high pedestrian density.

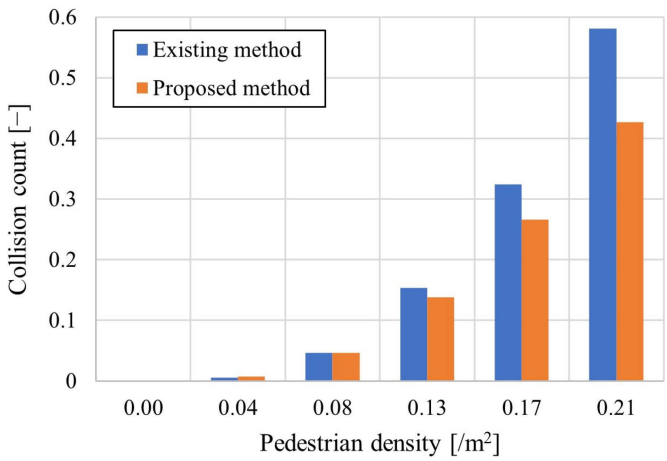


Figure 20. Comparison of the collision count among different pedestrian densities.

However, in low pedestrian conditions, such as with 20 pedestrians (pedestrian density $0.08/m^2$), the proposed method may result in a longer avoidance duration (Figure 15), a higher PS invasion ratio (Figure 17), and an increased travel time (Figure 18) compared to the existing method. In addition, as shown in Figure 20, under the condition of 10 pedestrians (pedestrian density $0.04/m^2$), the proposed method may result in a higher likelihood of collisions. Concerning the causes of these negative effects of the proposed method, the proposed method tolerates pedestrians approaching, even in situations with a sufficient margin around the PMV to freely perform avoidance behavior, and it could not initiate avoidance, such as turning in advance, as in the existing method. Consequently, a more pronounced avoidance behavior, such as suspending, was needed, leading to longer avoidance duration and overall travel time. Furthermore, it is considered that the more pronounced avoidance may cause the PMV to move in a direction that obstructs pedestrian flow, leading to an increase in the PS invasion ratio and the number of collisions. This change due to pedestrian density is considered one of the limitations of the proposed method. Based on these results, it is considered that the proposed method has a positive effect on the efficiency of autonomous navigation in high pedestrian density areas, such as conditions with a pedestrian density of $0.15/m^2$ or higher, rather than in low-density environments. Therefore, by observing pedestrian density and switching the path-planning method to the proposed method when the density is high, a more efficient autonomous navigation can be achieved without increasing passenger discomfort.

Additionally, although the effectiveness of the proposed method was indicated in several crowded environments targeted by the existing method [12], in a real-world environment, various factors exist, such as the shape of the passage and the characteristics of pedestrians, along with pedestrian density. For example, in environments where pedestrians are not organized and move randomly, unlike the simulation setting in this study, it may be challenging for the proposed method to achieve efficient autonomous navigation. Therefore, investigating how these environmental factors affect passenger tolerance, and the autonomous navigation is also one of the future challenges for the social implementation.

6. Conclusions

This study aimed to propose a path-planning method considering characteristics of passenger tolerance to approaching pedestrians in crowded environments to realize efficient autonomous navigation without increasing passenger discomfort. First, data collection and data analysis were done to clarify the characteristics of passenger tolerance. Next, considering the characteristics, a path-planning method that can achieve short travel time by suppressing avoidance behavior to tolerable pedestrians was proposed. Finally, the usefulness of the proposed method was evaluated based on numerical simulations under multiple pedestrian density conditions. As a result of the above, the following findings were obtained.

- The curvature radius of the pedestrian trajectory and passing distance between the pedestrian, predicted from the passenger's perspective, affect passenger tolerance. When the PMV does not perform avoidance behavior, the passenger can tolerate an approaching pedestrian if the passing distance is long, and the pedestrian's trajectory is straight.
- Considering the characteristics of passenger tolerance, the proposed path-planning method for autonomous navigation has the potential to reduce the travel time to reach the destination without increasing passenger discomfort. Furthermore, by selecting paths that do not obstruct pedestrian flow, the number of collisions with pedestrians can also be reduced, and safety can be improved.

This study conducted experiments using young healthy participants to understand basic passenger tolerance characteristics, minimizing the influence of differences in visual and cognitive performance. However, because visual acuity and cognitive function deteriorate with age, there may be differences between young and older passengers, who represent the primary user group of PMVs, in the way they feel discomfort and in their tolerance characteristics toward approaching pedestrians. Furthermore, while this study demonstrated that efficient autonomous navigation could be achieved by considering passenger tolerance in the targeted environments, various environmental factors, such as urban layouts and pedestrian densities, influence the path planning of the proposed method. For example, in environments with more complex shapes, such as corners, pedestrian movements are expected to be more dynamic compared to straight corridors. This could increase the likelihood of situations where the PMV needs to perform avoidance maneuvers for surrounding pedestrians. Thus, expanding the conditions for validation to consider various factors is essential to evaluate the range of applicability in real-world environments. Additionally, although the impact of the proposed method on passenger comfort was estimated with the PS invasion ratio in the simulation, it has not been directly assessed on how passengers feel about the proposed method. For autonomous navigation in real-world environments, addressing computational resource limitations is also essential. The proposed method accounts for the movement of all surrounding pedestrians. Then, if the process cannot keep up with considering all pedestrians in crowded scenarios, this issue should be addressed by narrowing the range of pedestrians recognized and considered in the path planning.

Thus, our future work for social implementation includes investigating passenger tolerance characteristics among a larger size of participants, involving older individuals, and examining the applicability of the proposed method through more experiments with different conditions considering various environmental factors in real-world environments while ensuring safety.

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Informed Consent Statement: The participants were provided with an explanation of the procedures, and informed consent was obtained, before the experiment.

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References

1. Consumer Affairs Agency. Investigation Report of Accident Causation Based on Article 23, Paragraph 1 of Consumer Safety Act. 2016. Available online: https://www.caa.go.jp/policies/council/csic/report/report_009/pdf/report_009_180129_0001.pdf (accessed on 18 December 2023).
2. Niijima, S.; Sasaki, Y.; Mizoguchi, H. Autonomous navigation of electric wheelchairs in urban areas on the basis of self-generated 2D drivable maps. In Proceedings of the 2018 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM), Auckland, New Zealand, 9–12 July 2018; pp. 1081–1086. [CrossRef]
3. Mulky, R.S.; Koganti, S.; Shahi, S.; Liu, K. Autonomous scooter navigation for people with mobility challenges. In Proceedings of the 2018 IEEE International Conference on Cognitive Computing (ICCC), San Francisco, CA, USA, 2–7 July 2018; pp. 87–90. [CrossRef]
4. Jung, Y.; Kim, Y.; Lee, W.H.; Bang, M.S.; Kim, Y.; Kim, S. Path planning algorithm for an autonomous electric wheelchair in hospitals. *IEEE Access* **2020**, *8*, 208199–208213. [CrossRef]
5. Park, J.J.; Johnson, C.; Kuipers, B. Robot navigation with model predictive equilibrium point control. In Proceedings of the 2012 IEEE/RSJ International Conference on Intelligent Robots and Systems, Vilamoura-Algarve, Portugal, 7–12 October 2012; pp. 4945–4952. [CrossRef]
6. Trautman, P.; Krause, A. Unfreezing the robot: Navigation in dense, interacting crowds. In Proceedings of the 2010 IEEE/RSJ International Conference on Intelligent Robots and Systems, Taipei, Taiwan, 18–22 October 2010; pp. 797–803. [CrossRef]
7. Schockenhoff, F.; Nehse, H.; Lienkamp, M. Maneuver-based objectification of user comfort affecting aspects of driving style of autonomous vehicle concepts. *Appl. Sci.* **2020**, *10*, 3946. [CrossRef]
8. Hall, E.T. *The Hidden Dimension*; Doubleday: Garden City, NY, USA, 1966.
9. Pham, T.Q.; Nakagawa, C.; Shintani, A.; Ito, T. Evaluation of the effects of a personal mobility vehicle on multiple pedestrians using personal space. *IEEE Trans. Intell. Transp. Syst.* **2015**, *16*, 2028–2037. [CrossRef]
10. Toyoshima, A.; Nishino, N.; Chugo, D.; Muramatsu, S.; Yokota, S.; Hashimoto, H. Autonomous mobile robot navigation: Consideration of the pedestrian's dynamic personal space. In Proceedings of the 2018 IEEE 27th International Symposium on Industrial Electronics (ISIE), Cairns, QLD, Australia, 13–15 June 2018; pp. 1094–1099. [CrossRef]
11. Harada, R.; Yoshitake, H.; Shino, M. Pedestrian's avoidance behavior characteristics against autonomous personal mobility vehicles for smooth avoidance. *J. Robot. Mechatron.* **2024**, *36*, 918–927. [CrossRef]
12. Isono, Y.; Yoshitake, H.; Shino, M. Autonomous navigation method considering passenger comfort recognition for personal mobility vehicles in crowded pedestrian spaces. In Proceedings of the International Joint Conference on Computer Vision, Imaging and Computer Graphics, Virtual Event, 6–8 February 2023; pp. 182–202. [CrossRef]
13. Endsley, M.R. Toward a theory of situation awareness in dynamic systems. *Hum. Factors* **1995**, *37*, 32–64. [CrossRef]
14. Yoshitake, H.; Isono, Y.; Shino, M. Pedestrian avoidance method considering passenger comfort for autonomous personal mobility vehicles. *J. Robot. Mechatron.* **2023**, *35*, 231–239. [CrossRef]
15. Zanlungo, F.; Ikeda, T.; Kanda, T. Social force model with explicit collision prediction. *Europhys. Lett.* **2011**, *93*, 68005. [CrossRef]

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